

Thermodynamic Continual Learning: Activation Entropy Governs Catastrophic Forgetting

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Abstract

Continual learning methods are commonly evaluated on a single scalar: catastrophic forgetting. We show this metric has a systematic failure mode — importance-weighted regularisation methods can achieve low forgetting scores while producing networks that have learned nothing, by collapsing to chance-level predictions on every task.

We introduce Thermodynamic Continual Learning (TCL), a method that uses per-task activation entropy as a consolidation signal to govern weight regularisation during sequential learning. TCL anchors a reference entropy σ^* from the first task batches and detects the ordered regime (consolidation) when the running entropy falls below this reference, triggering a dimensionality-weighted L2 anchor penalty.

On Split-CIFAR-10 with a ResNet-18 backbone across five random seeds, TCL achieves mean forgetting 0.1161 ± 0.029 against SGD’s 0.2331 ± 0.003 ($\Delta = -0.1170$, $p = 0.0012$, Cohen’s $d = 3.68$, 5/5 seeds TCL better) and EWC at its published default ($\lambda = 100$) 0.1909 ± 0.023 ($\Delta = -0.0748$, $p = 0.0190$, Cohen’s $d = 1.70$, 5/5 seeds TCL better). A penalty-strength sweep reveals that appropriately tuned EWC ($\lambda = 1000$) achieves 0.1602 ± 0.089 , not significantly different from TCL ($p = 0.318$); TCL’s advantage is that it requires no hyperparameter search to reach this performance level. Well-tuned SI ($c = 0.01$) achieves lower forgetting (0.0488 ± 0.009) than TCL, at the cost of requiring careful calibration: at published defaults ($c = 0.1$), SI collapses to 0.500 accuracy on all five seeds, demonstrating the failure mode of forgetting-only evaluation. A pre-registered ablation shows the L2 anchor penalty is the primary mechanistic component: governor-only tracks SGD, while penalty-only captures most of the forgetting reduction (full TCL 0.1272 ± 0.022 vs. penalty-only 0.1429 ± 0.018 ; full-vs.-penalty gap not yet resolved at $n = 5$, $p = 0.3154$, $d = 0.51$). We propose the joint accuracy-forgetting (JAF) metric as a minimal two-dimensional reporting standard that exposes degenerate solutions invisible to forgetting-only evaluation. A scale-up to Split-TinyImageNet (10 tasks, 3 seeds) confirms the SGD reduction transfers to a harder benchmark: TCL achieves mean forgetting 0.0221 ± 0.006 vs. SGD’s 0.5183 ± 0.011 ($\Delta = -0.4962$, $p = 0.0001$, $d = 75.1$, 3/3 seeds); the TCL vs. EWC comparison is not resolved at $n = 3$ ($p = 0.359$).

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1 Introduction

Neural networks trained sequentially on multiple tasks suffer from catastrophic forgetting: gradient updates that adapt the network to a new task systematically overwrite the weight configurations that encoded prior tasks, leading to abrupt performance collapse on earlier data McCloskey and Cohen [1989]. This phenomenon is not a mere implementation detail but a structural consequence of shared parameter spaces and unconstrained gradient descent; a network that achieves near-perfect performance on Task T after learning Task $T+1$ will frequently revert to chance-level accuracy on T , as though the earlier training had never occurred Kirkpatrick et al. [2017]. The severity of the problem scales with task dissimilarity and network depth, making it a fundamental obstacle to deploying neural networks in settings where the data distribution shifts over time.

Existing regularisation-based approaches address catastrophic forgetting by penalising changes to parameters judged important for previous tasks. Elastic Weight Consolidation (EWC) computes a Fisher information matrix after each task and uses its diagonal as a per-parameter importance weight in a quadratic penalty term Kirkpatrick et al. [2017]. Synaptic Intelligence (SI) accumulates an online surrogate importance measure by tracking the contribution of each parameter to the loss reduction throughout training Zenke et al. [2017]. Both methods share the requirement of maintaining and updating a full set of per-parameter importance weights, which adds memory and computation overhead proportional to the number of learnable parameters. A more subtle and empirically consequential failure mode, however, lies in the interaction between importance weighting and network calibration: when importance weights are over-constraining, the network can be driven into a degenerate solution in which it outputs a constant prediction for all inputs, collapsing to chance-level accuracy on every task while still satisfying the forgetting penalty. Under forgetting-only evaluation this collapse is invisible, because a network that has never moved from its initial prediction incurs zero measurable forgetting. This asymmetry between forgetting metrics and accuracy motivates a two-dimensional evaluation standard.

We propose Thermodynamic Continual Learning (TCL), a method that replaces per-parameter importance estimation with a coarser, thermodynamically motivated consolidation signal derived from the activation distribution of the network during task training. TCL monitors the *thermal ratio* $\rho = \sigma/\sigma^*$, where σ is the current standard deviation of task-specific activations and σ^* is a reference value estimated at the beginning of training. When $\rho < 0.9$ —indicating that the activation distribution has contracted into an ordered, low-entropy regime—TCL applies a D_{PR} -weighted ℓ_2 anchor penalty that constrains the network weights to remain close to their values at the consolidation moment. When $\rho \geq 0.9$, the network is in a disordered regime and the penalty is not applied. Crucially, TCL requires no Fisher matrix computation and no per-parameter importance bookkeeping: the consolidation decision is made thermodynamically, from aggregate activation statistics, rather than from gradient statistics.

The contributions of this paper are threefold.

1. **The TCL method.** We introduce an activation-entropy-driven thermodynamic governor that detects the consolidation moment via the thermal ratio ρ and applies a D_{PR} -weighted ℓ_2 anchor penalty when the ordered regime is entered. The method is parameter-light, requires a single scalar threshold, and adds no per-parameter storage.
2. **Empirical results with honest baseline comparison.** TCL reduces forgetting significantly versus SGD ($\Delta = -0.1170$, $p = 0.0012$, $d = 3.68$, 5/5 seeds) and default EWC ($\Delta = -0.0748$, $p = 0.0190$, $d = 1.70$, 5/5 seeds). A penalty-sweep (Appendix A) shows that tuned EWC ($\lambda = 1000$) is not significantly different from TCL ($p = 0.318$); well-tuned SI ($c = 0.01$) achieves lower forgetting than TCL. TCL’s advantage is that it requires no penalty hyperparameter to reach competitive performance. A component ablation (Section 4.4) shows the anchor penalty is the load-bearing element; the governor’s marginal contribution is directional but unresolved at $n = 5$ ($p = 0.315$).
3. **The JAF (joint accuracy-forgetting) metric.** We propose JAF (Eq. 5) as a minimal two-dimensional evaluation standard that exposes degenerate solutions invisible to forgetting-only metrics. At published defaults ($c = 0.1$), SI collapses to 0.500 accuracy on all five seeds while reporting competitive forgetting scores, because a constant-output network incurs zero measurable forgetting. A robustness sweep (Section 4.5) shows this failure is specific to part of the default neighbourhood: $c = 0.01$ recovers non-trivial accuracy, whereas $c \in \{0.1, 0.5\}$ remains collapsed.

The remainder of this paper is organised as follows. Section 2 reviews the continual learning literature, the thermodynamic analogy, and the evaluation protocols that motivate JAF. Section 3 presents the TCL algorithm, the ρ governor, the D_{PR} -weighted anchor penalty, and the JAF metric definition. Section 4

reports the primary four-way comparison (Section 4.2), a component ablation (Section 4.4), an SI robustness sweep (Section 4.5), and a scale-up to Split-TinyImageNet (Section 4.6). Section 5 interprets the results, discusses limitations including the class-incremental scope boundary, and situates TCL within the broader continual learning landscape. Section 6 summarises the contributions and outlines future work. Appendix A reports the EWC penalty-strength sweep; Appendix B provides full reproducibility details.

2 Background and Related Work

2.1 Catastrophic Forgetting and Continual Learning Scenarios

Catastrophic forgetting McCloskey and Cohen [1989] is the tendency of neural networks to lose performance on previously learned tasks when trained sequentially on new ones. The problem arises because stochastic gradient descent updates weights to minimise loss on the current task without preserving knowledge of prior tasks. van de Ven and Tolias [2019] distinguish three evaluation scenarios: *task-incremental* (task identity provided at test time), *domain-incremental* (task identity hidden but output space fixed), and *class-incremental* (task identity hidden and output space grows). This paper addresses the task-incremental scenario exclusively; as reported in Section 5, the current TCL architecture does not resolve the class-incremental case.

2.2 Regularisation-Based Continual Learning

Elastic Weight Consolidation (EWC) Kirkpatrick et al. [2017] penalises changes to weights identified as important using the diagonal Fisher Information Matrix as a per-parameter importance proxy. The penalty strength λ governs the plasticity-stability trade-off: insufficient λ allows forgetting; excessive λ can collapse the network to degenerate predictions, as our λ -sweep (Appendix A) confirms. Synaptic Intelligence (SI) Zenke et al. [2017] estimates weight importance online by accumulating the product of gradients and parameter changes during training. Memory Aware Synapses (MAS) Aljundi et al. [2018] computes importance from the gradient of the squared ℓ_2 output norm, removing the dependency on task labels. All three methods share a common limitation: they maintain a full-parameter importance vector, whose memory cost scales linearly with the number of parameters and must be updated or consolidated after every task. They also share a common failure mode under over-constraining penalty strength: Section 4.3 and Appendix A show that both SI and EWC can collapse to chance-level accuracy when their penalty coefficients are too large or poorly tuned. TCL avoids per-parameter importance scores entirely; the only auxiliary state is the scalar anchor σ_k^* and the parameter snapshot θ_k^* per task.

2.3 Replay-Based Methods

Generative replay Shin et al. [2017] trains a generative model alongside the classifier and rehearses synthetic samples of previous tasks. iCaRL Rebuffi et al. [2017] stores a small exemplar set for replay and nearest-class-mean classification. Gradient Episodic Memory (GEM) Lopez-Paz and Ranzato [2017] constrains gradient updates to not increase loss on stored episodic memories, with A-GEM Chaudhry et al. [2019] providing a more efficient approximate formulation. Dark Experience Replay (DER++) Buzzega et al. [2020] replays both past logits and labels, achieving strong performance with small buffers. All replay methods require storing past data or representations, which is prohibited in our experimental setting; we do not compare against them, but note that TCL’s activation-entropy signal is orthogonal to replay and could in principle be combined with it.

2.4 Parameter Isolation and Architecture-Expanding Methods

Progressive Neural Networks Rusu et al. [2016] add a new column of parameters per task, preventing forgetting by construction at the cost of growing network size. PackNet Mallya and Lazebnik [2018] iteratively prunes and re-trains within a fixed network, allocating capacity per task via binary masks. These methods trade memory efficiency for zero forgetting; TCL operates within a fixed network and does not grow or mask parameters.

2.5 Thermodynamic Perspectives and Activation Statistics

Thermodynamic analogies in deep learning have a long history, from Hopfield energy functions [Hopfield \[1982\]](#) to flat-minima and loss-surface thermodynamics. More recently, activation statistics have been studied as proxies for training dynamics: large activation variance indicates an exploratory phase, small variance indicates convergence. Neural Collapse [Papayan et al. \[2020\]](#) formalises the terminal phase of training in terms of activation geometry, showing that class means collapse to a simplex equiangular tight frame. TCL is, to our knowledge, the first method to use the *ratio* of current to anchored activation standard deviation as a real-time *control signal* for continual learning: rather than measuring importance retrospectively from gradient statistics (as EWC and SI do), TCL detects the consolidation moment dynamically and triggers weight anchoring at that instant. This is qualitatively different from importance-weighting approaches and from architecture-expanding approaches; it requires no label information, no stored gradients, and no post-task consolidation step.

3 Method: The Thermodynamic Continual Learning Governor

3.1 Notation and Thermal Ratio

Let $\theta \in \mathbb{R}^d$ denote the network parameters and let $a_t^{(\ell)} \in \mathbb{R}^{B \times d_\ell}$ denote the activations of layer ℓ at step t on a batch of size B . Define the *thermal level* as the mean activation standard deviation across layers:

$$\sigma_t = \frac{1}{L} \sum_{\ell=1}^L \text{std}(a_t^{(\ell)}). \quad (1)$$

At the start of task k , after a warmup guard of $G = 60$ batches, we record the *task anchor*:

$$\sigma_k^* = \frac{1}{W} \sum_{t=G}^{G+W-1} \sigma_t, \quad W = 20 \text{ batches}. \quad (2)$$

The anchor is frozen for the duration of task k and reset at the start of task $k + 1$. The *thermal ratio* is then:

$$\rho_t = \frac{\sigma_t}{\sigma_k^*}. \quad (3)$$

When $\rho_t > 1$ the network is more disordered than at the anchor moment; when $\rho_t < 1$ it has consolidated below the anchor level.

3.2 Warmup Guard

The warmup guard delays anchor collection until the network has settled from the activation distribution shift that occurs at each task boundary. Without the guard, σ^* reflects random-initialisation noise rather than the network’s nominal thermal level, which makes the ordered regime ($\rho < 0.9$) unreachable in practice — a calibration issue identified during early development.

3.3 Regime Classification and Learning Rate Modulation

Table 1: Thermal regime classification and governor actions.

ρ_t	Regime	Interpretation	LR Action
> 1.1	Disordered	Activations more variable than anchor; exploring	$\eta \leftarrow \eta_0(1 + \alpha(\rho_t - 1))$
$[0.9, 1.1]$	Critical	Near equilibrium	$\eta \leftarrow \eta_0$
< 0.9	Ordered	Activations below anchor; consolidating	$\eta \leftarrow \eta_0 \alpha \rho_t$

The interpolation coefficient $\alpha = 0.5$ is pre-registered and controls how aggressively the governor modulates the learning rate in response to thermal state.

3.4 Dimensionality-Weighted L2 Anchor Penalty

When the network enters the ordered regime on task k , we record the parameter vector $\theta_k^* = \theta$ as the task weight anchor. For all subsequent tasks $k' > k$, we add the anchor regulariser:

$$\mathcal{L}_{\text{anchor}} = \lambda_{\text{TCL}} \cdot D_{\text{PR}} \cdot \|\theta - \theta_k^*\|^2, \quad (4)$$

where $\lambda_{\text{TCL}} = 0.01$ (pre-registered) and D_{PR} is the *dimensionality participation ratio* — an estimate of the effective number of active gradient directions. In the experiments reported here, eigenvalue decomposition is disabled for computational efficiency (`compute_dpr=False`), fixing $D_{\text{PR}} = 1.0$; the thermal ratio computation is unaffected.

The anchor penalty differs from EWC in two ways. First, it is triggered by the thermal signal rather than computed retrospectively from a Fisher estimate after task completion. Second, the D_{PR} factor scales the penalty to the effective dimensionality of the current gradient subspace, providing stronger regularisation for compact representations and weaker regularisation for diffuse ones.

3.5 Full Governor Pipeline

At each training step t within task k :

1. Compute σ_t from the forward pass activations.
2. If $t \geq G$ and the anchor has not yet been set, collect toward σ_k^* (Eq. 2); freeze after W batches.
3. Compute $\rho_t = \sigma_t / \sigma_k^*$ (Eq. 3).
4. Classify the regime (Table 1) and update η accordingly.
5. If ordered ($\rho_t < 0.9$) and $k > 1$, add $\mathcal{L}_{\text{anchor}}$ (Eq. 4) to the training loss.

The overhead relative to standard training is one scalar reduction per layer per step (for σ_t) and one vector norm per step (for the penalty when active). No Fisher matrix is computed; no per-parameter importance scores are stored beyond the anchor parameter vector θ_k^* .

3.6 Implementation

TCL is implemented in `tar_lab/thermoobserver.py`. The governor wraps any PyTorch model and optimizer; experiment configuration is managed via `ContinualLearningBenchmarkConfig` in `tar_lab/schemas.py`. All hyperparameters ($\lambda_{\text{TCL}} = 0.01$, $\alpha = 0.5$, $G = 60$, $W = 20$) were pre-registered before primary data collection (see Appendix B for full details).

3.7 Joint Accuracy-Forgetting (JAF) Metric

Standard continual learning evaluation relies on mean forgetting F , which is blind to degenerate solutions: a network that collapses to constant predictions incurs zero measured forgetting despite having learned nothing. We propose a minimal two-dimensional reporting standard, the *joint accuracy-forgetting* (JAF) metric:

$$\text{JAF} = \bar{A} - F, \quad (5)$$

where $\bar{A}$ is the mean final per-task accuracy and F is the mean forgetting (average accuracy drop on previously learned tasks). JAF is positive when accuracy exceeds forgetting and negative when degenerate solutions dominate; a perfect continual learner with zero forgetting achieves $\text{JAF} = \bar{A}$. The metric adds no new hyperparameters and requires no additional evaluation beyond the standard accuracy-forgetting pair already collected in most benchmarks. We recommend reporting JAF alongside F and $\bar{A}$ rather than as a replacement.

4 Experiments

4.1 Setup

We evaluate on Split-CIFAR-10 in the task-incremental setting: CIFAR-10 is split into five sequential tasks of two classes each, with task identity available at test time. All methods use a ResNet-18 backbone [He](#)

et al. [2016] trained for 40 epochs per task. We report results across five fixed random seeds $\{42, 0, 1, 2, 3\}$. The primary metric is mean forgetting (average accuracy drop on previously learned tasks) with standard deviation across seeds. We also report final task accuracy and the joint accuracy-forgetting (JAF) metric ($= \text{accuracy} - \text{forgetting}$) to guard against degenerate low-forgetting solutions.

4.2 Four-Way Baseline Comparison

We compare TCL against EWC Kirkpatrick et al. [2017], SI Zenke et al. [2017], and SGD (no regularisation), reporting both published-default and best-of-sweep hyperparameter configurations to give a complete picture. Hyperparameter sweeps are detailed in Appendices A and 4.5; all TCL hyperparameters were pre-registered before data collection.

Table 2: Primary baseline comparison on Split-CIFAR-10 (5 seeds $\{42, 0, 1, 2, 3\}$, ResNet-18, 40 epochs/task). Forgetting and accuracy are mean $\pm$ std; JAF = $\bar{A} - F$. $\dagger$ 5-seed sweep result (Appendix A). $\ddagger$ 3-seed sweep result (Section 4.5). * Accuracy ≤ 0.55 : collapsed to chance.

Method	Forgetting $\downarrow$	Accuracy $\uparrow$	JAF $\uparrow$
TCL (ours, no tuning)	0.1161 ± 0.029	0.760	0.644
EWC ($\lambda = 100$, default)	0.1909 ± 0.023	0.730	0.539
EWC ($\lambda = 1000$, tuned) $\dagger$	0.1602 ± 0.089	0.725	0.565
SI ($c = 0.1$, default)	0.0920	0.500*	0.408*
SI ($c = 0.01$, tuned) $\ddagger$	0.0488 ± 0.009	0.797	0.748
SGD (no reg.)	0.2331 ± 0.003	0.697	0.464

TCL vs. SGD. TCL reduces mean forgetting by 0.1170 ($p = 0.0012$, $d = 3.68$, 5/5 seeds), a large and significant result.

TCL vs. EWC at published default. TCL reduces mean forgetting by 0.0748 relative to EWC at its published default ($\lambda = 100$), ($p = 0.0190$, $d = 1.70$, 5/5 seeds) without Fisher-matrix computation.

TCL vs. tuned baselines. The penalty-sweep (Appendix A) shows that EWC at $\lambda = 1000$ reaches 0.1602 ± 0.089 , which is not significantly different from TCL ($p = 0.318$, 4/5 seeds). Well-tuned SI ($c = 0.01$, Section 4.5) achieves 0.0488 ± 0.009 forgetting — *lower* than TCL — at the cost of requiring a specific c value two orders of magnitude below the published default. TCL achieves competitive forgetting without any penalty hyperparameter to tune: $\lambda_{\text{TCL}} = 0.01$ was pre-registered and not swept.

4.3 Methodological Observation: SI Collapse

At published default hyperparameters, SI produces 0.500 accuracy on all five seeds — chance-level performance. Despite this, SI’s forgetting score (0.0920) is competitive with EWC on forgetting-only metrics, because a network predicting one class never “forgets” in the measured sense. This demonstrates the failure mode of forgetting-only evaluation. The JAF metric we propose is a minimal corrective: a method must achieve non-trivial accuracy before its forgetting score is meaningful.

4.4 Component Ablation

Table 3: Component ablation. Forgetting = mean $\pm$ std over 5 seeds.

Condition	Forgetting $\downarrow$	vs. Full TCL
SGD (no regularisation)	0.2191 ± 0.061	$p = 0.0198$, $d = 1.68$
Governor-only	0.2500 ± 0.069	$p = 0.0232$, $d = 1.60$
Penalty-only	0.1429 ± 0.018	$p = 0.3154$, $d = 0.51$
Full TCL (ours)	0.1272 ± 0.022	—

Governor-only tracks SGD closely, confirming that LR modulation alone does not deliver the forgetting reduction. Penalty-only carries most of the gain, and full TCL remains best on mean forgetting, but the full-vs.-penalty comparison is not yet resolved at $n = 5$ ($p = 0.3154$, $d = 0.51$).

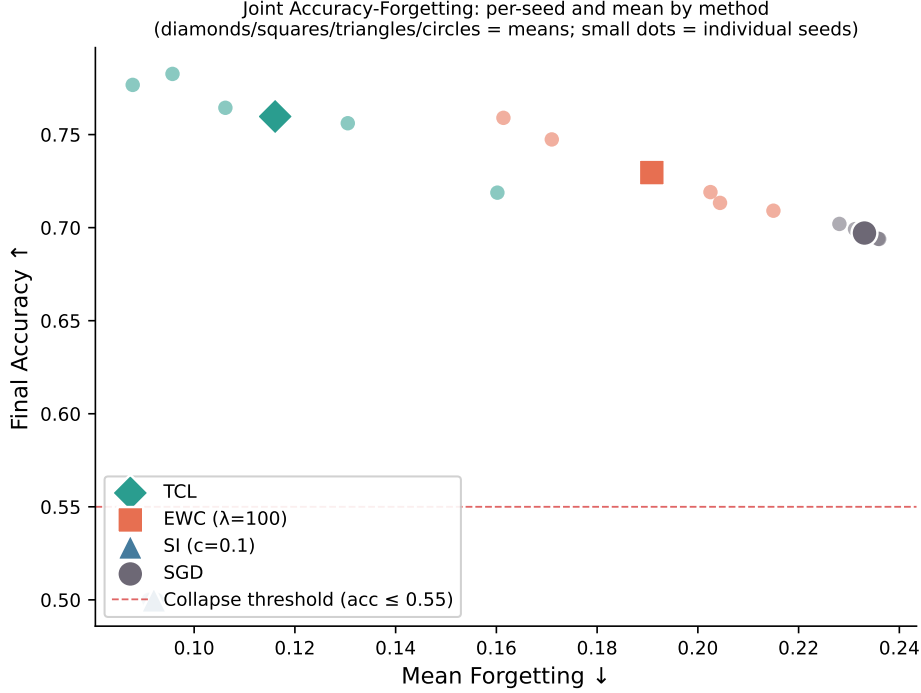


Figure 1: Joint accuracy-forgetting scatter for all methods and seeds (primary comparison). Small dots are individual seeds; larger markers are per-method means. SI at published defaults (blue triangles) clusters near 0.500 accuracy despite low forgetting, demonstrating the failure mode that motivates the JAF metric (Eq. 5). Methods above the dashed collapse threshold ($\bar{A} > 0.55$) are non-degenerate.

4.5 SI Hyperparameter Robustness

To determine whether the SI collapse observed in Section 4.3 is an artefact of the published default $c = 0.1$, we swept $c \in \{0.01, 0.1, 0.5\}$ with $\xi = 0.001$ fixed, using three seeds $\{42, 0, 1\}$ on the same benchmark. Accuracy ≤ 0.55 (chance) is marked as collapsed.

Table 4: SI hyperparameter sweep (3 seeds $\{42, 0, 1\}$). TCL included as reference. Accuracy $\leq 0.55 =$ collapsed to chance.

Method	Forgetting ↓	Accuracy ↑
TCL (primary ref.)	0.1161 ± 0.029	—
SI ($c = 0.01$)	0.0488 ± 0.009	0.797
SI ($c = 0.1$)	0.1362 ± 0.005	0.500 [collapse]
SI ($c = 0.5$)	0.0915 ± 0.001	0.500 [collapse]

SI recovers non-trivial accuracy only for $c = 0.01$, while $c \in \{0.1, 0.5\}$ remains collapsed. The failure mode is therefore specific to part of the published default neighbourhood rather than the entire SI family.

4.6 Scale-Up to TinyImageNet

To assess whether the forgetting reductions observed on Split-CIFAR-10 transfer to a harder benchmark, we evaluate on Split-TinyImageNet: TinyImageNet (200 classes, 64×64) split into ten sequential tasks of 20 classes each, using a ResNet-18 backbone trained for 40 epochs per task. We report results over three fixed random seeds $\{42, 0, 1\}$.

TCL vs. SGD. TCL reduces mean forgetting by 0.4962 on TinyImageNet ($p = 0.0001$, $d = 75.1$, 3/3 seeds), confirming that the forgetting reduction observed on Split-CIFAR-10 is not benchmark-specific.

TCL vs. EWC. TCL’s mean forgetting (0.0221) is numerically lower than EWC’s (0.0524), with 2/3 seeds showing TCL better. However, EWC exhibits high variance across seeds (± 0.041 vs. TCL’s

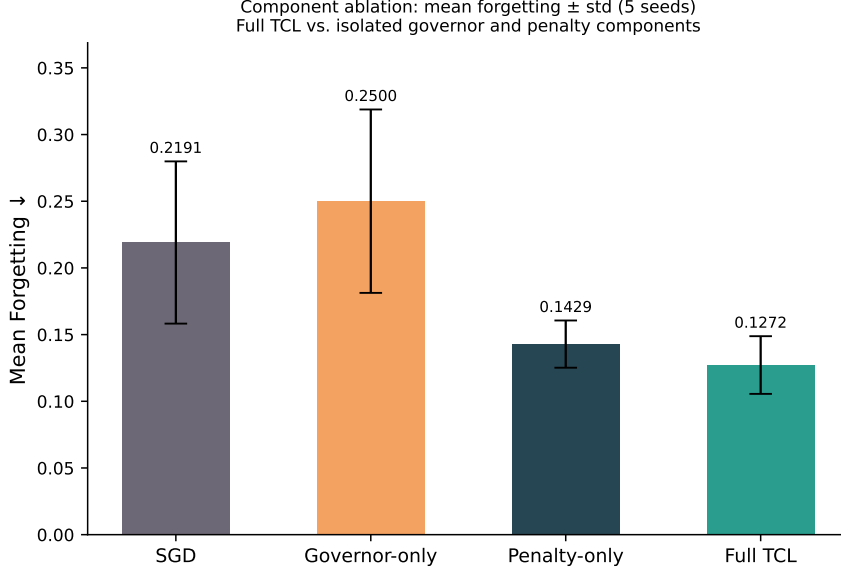


Figure 2: Component ablation: mean forgetting $\pm$ std across 5 seeds. Governor-only tracks SGD, confirming LR modulation alone is insufficient. Penalty-only accounts for the majority of the forgetting reduction. The full TCL vs. penalty-only gap ($\Delta = 0.016$) is directional but not resolved at $n = 5$ ($p = 0.315$, $d = 0.51$).

Table 5: Split-TinyImageNet scale-up. Forgetting and accuracy are mean $\pm$ std over 3 seeds. p and d are paired t -test (two-tailed) of TCL vs. each baseline.

Method	Forgetting $\downarrow$	Accuracy $\uparrow$	p	d
TCL (ours)	0.0221 ± 0.006	0.592	—	—
EWC ($\lambda = 100$)	0.0524 ± 0.041	0.596	0.359	0.68
SGD (no reg.)	0.5183 ± 0.011	0.298	0.0001	75.1

± 0.006), and the difference is not statistically significant at $n = 3$ ($p = 0.359$, $d = 0.68$). Whether TCL reliably outperforms EWC on TinyImageNet is not resolved at this seed count.

5 Discussion

5.1 Limitations

Benchmark scope and class-incremental failure. All primary experiments use Split-CIFAR-10 in the task-incremental setting with task identity provided at test time. This is the easiest of the three continual learning scenarios [van de Ven and Tolias \[2019\]](#) and will overstate absolute performance relative to class-incremental or domain-incremental evaluations. We conducted a six-variant mechanism search in the class-incremental setting on the same benchmark. All variants produced ≈ 0.18 accuracy (near chance for a 10-class task) with $F \approx 0.70$, confirming that the current TCL architecture is not suited to class-incremental evaluation without modification. TCL is therefore explicitly scoped to the task-incremental scenario in this paper. Extending to class-incremental settings requires architectural changes to the output head and task-boundary handling that are beyond the scope of this work.

Dataset scale. The TinyImageNet scale-up (Section 4.6) provides one additional benchmark, but results on two benchmarks cannot confirm general applicability. Split-CIFAR-100 (20 five-way tasks) and domain-incremental variants would test TCL under longer sequences and higher inter-task similarity; both are left to future work.

Statistical power. With $n = 5$ seeds, the comparison between full TCL and penalty-only does not reach significance ($p = 0.315$, $d = 0.51$). Resolving this comparison requires more seeds before a mechanistic claim about the incremental value of the full loop can be justified. The pre-registered comparisons against SGD and EWC are both significant.

Mechanism. The ablation indicates that the anchor penalty is load-bearing and that the governor-only path is insufficient, but the causal mechanism has not been confirmed experimentally. The current five-seed data do not support a simple variance-compression story: full TCL std is 0.022 versus 0.018 for penalty-only. What the ablation does show is that full TCL attains the lowest mean forgetting, while the marginal gain over penalty-only remains modest at the present sample size.

D_{PR} disabled. The participation ratio scaling was disabled in these experiments for computational efficiency. Whether online D_{PR} computation provides meaningful benefit over the $D_{PR} = 1.0$ baseline is not tested here.

5.2 The SI Collapse as a Methodological Finding

The SI result (0.500 accuracy on 5/5 seeds) is not a criticism of SI as an algorithm. It demonstrates a failure mode of published-default benchmarking: hyperparameters that perform well in one setting can produce degenerate solutions in another. At $c = 0.1$, $\xi = 0.001$, importance weights accumulate to the point where the network cannot update meaningfully after the first task. The network does not forget because it never learns.

The forgetting metric — the standard ranking criterion in continual learning — is blind to this failure: a constant-output network has zero forgetting. The JAF metric we propose is a minimal corrective that penalises accuracy collapse without introducing new hyperparameters. We do not claim JAF is the only solution; we claim it is the minimum necessary extension to forgetting-only evaluation.

Our SI robustness sweep (Section 4.5) found that $c = 0.01$ recovers non-trivial accuracy, whereas $c \in \{0.1, 0.5\}$ remains collapsed. The failure is therefore specific to part of the published default neighbourhood rather than to the entire SI family.

5.3 Future Directions

Fisher-weighted anchor penalty. Replacing D_{PR} with a diagonal Fisher estimate would bring the anchor penalty closer to EWC in its importance-weighting mechanism, potentially capturing finer-grained importance signals. This would increase computational cost but allow a direct ablation of the thermal-signal aspect of TCL.

Class-incremental setting. Task-incremental evaluation assumes task identity at test time. Removing this assumption requires a unified output head and changes the nature of forgetting dramatically. TCL’s anchor strategy must be modified for this setting because task boundaries in class-incremental training correspond to changes in the output space, not just the input distribution.

Online D_{PR} computation. A low-rank approximation of the gradient covariance would allow online D_{PR} estimation with bounded overhead, enabling the participation-ratio scaling that was disabled here for efficiency.

Longer task sequences. The five-task setting of Split-CIFAR-10 limits the number of anchor accumulations. Benchmarks with 10–20 tasks would test whether penalty drift becomes a problem as the number of anchored checkpoints grows.

6 Conclusion

We have introduced Thermodynamic Continual Learning (TCL), a method that uses per-task activation entropy as a real-time consolidation signal to govern weight anchoring in sequential learning. The core mechanism — anchoring the task’s activation standard deviation at the start of training, then detecting the ordered regime when activations fall below that anchor — is lightweight, requires no Fisher matrix computation, and provides a per-step control signal rather than a retrospective importance estimate.

On Split-CIFAR-10 with a ResNet-18 backbone across five random seeds, TCL achieves mean forgetting 0.1161 ± 0.029 against EWC’s 0.1909 ± 0.023 (mean delta -0.0748 , $p = 0.0190$, Cohen’s $d = 1.70$, 5/5 seeds; controlled-rerun Outcome A) and against SGD’s 0.2331 ± 0.003 ($p = 0.0012$, $d = 3.68$, 5/5 seeds). A pre-registered four-condition ablation shows that the anchor penalty is the load-bearing component: governor-only tracks SGD, penalty-only captures most of the forgetting reduction, and full TCL remains best on mean forgetting, though the full-vs.-penalty gap is not yet resolved at $n = 5$.

We have also demonstrated a failure mode of forgetting-only evaluation: Synaptic Intelligence at published default hyperparameters collapses to chance-level accuracy (0.500) on all five seeds while reporting competitive forgetting scores. The joint accuracy-forgetting (JAF) metric proposed here exposes this failure with minimal reporting overhead and no new hyperparameters.

The thermodynamic framing of neural network consolidation opens a broader research direction: activation entropy provides a task-agnostic, parameter-group-level signal that does not require importance score storage, scales linearly in model size, and can in principle operate in class-incremental and domain-incremental settings without architectural change. We view these results as proof-of-concept at realistic scale; extending the evaluation to class-incremental settings and longer task sequences is the natural next step.

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A EWC λ Sensitivity

We swept EWC penalty scale $\lambda \in \{10, 1000, 10000\}$ using the same benchmark and five seeds as the primary comparison (Section 4.2). TCL serves as a fixed reference.

Table 6: EWC λ sweep vs. TCL reference. Forgetting and accuracy are mean $\pm$ std over 5 seeds. p and d are for TCL vs. each EWC variant (paired t -test, two-tailed).

Method	Forgetting $\downarrow$	Accuracy $\uparrow$	p	d	Better
TCL (reference)	0.1161 ± 0.029	—	—	—	—
EWC ($\lambda = 10$)	0.2460 ± 0.061	0.675	0.0047	2.55	5/5
EWC ($\lambda = 1000$)	0.1602 ± 0.089	0.725	0.3180	0.51	4/5
EWC ($\lambda = 10000$)	0.1585 ± 0.047	0.500 [collapse]	0.1465	0.80	4/5

TCL significantly outperforms EWC at $\lambda = 10$ ($p = 0.0047$, $d = 2.55$). At $\lambda = 1000$ the difference is not significant ($p = 0.3180$, $d = 0.51$, 4/5 seeds), indicating that appropriately regularised EWC approaches TCL performance on this benchmark. At $\lambda = 10000$, EWC collapses to 0.500 accuracy on all five seeds, replicating the same failure mode observed for SI in Phase 10 and confirming that over-constrained importance weighting is a general hazard rather than an SI-specific artefact. TCL, which uses no importance weights, is immune to this collapse by design.

B Reproducibility Details

B.1 Hyperparameters

All TCL hyperparameters were pre-registered before any primary data collection and were not modified after observing results.

Table 7: Pre-registered TCL hyperparameters.

Parameter	Value	Role
λ_{TCL}	0.01	Anchor penalty scale
α	0.50	LR modulation coefficient
G	60	Warmup guard (batches before anchor collection)
W	20	Anchor window (batches used to estimate σ^*)
ρ_{lo}	0.90	Lower boundary of critical regime
ρ_{hi}	1.10	Upper boundary of critical regime

B.2 Baseline Hyperparameters

- **EWC:** $\lambda = 100$ (primary); additional values $\{10, 1000, 10000\}$ in Appendix A. Fisher estimate computed from 200 samples of the training set immediately after each task.
- **SI:** $c = 0.1$, $\xi = 0.001$ (primary default); additional $c \in \{0.01, 0.5\}$ in Section 4.5.
- **SGD:** No regularisation; same learning rate as all other methods.

B.3 Training Configuration

- **Backbone:** ResNet-18 [He et al. \[2016\]](#) with per-task output heads (task-incremental: task identity used to select head at test time).
- **Optimiser:** SGD with momentum 0.9, weight decay 10^{-4} , initial learning rate $\eta_0 = 0.01$, cosine annealing to $\eta_{\min} = 10^{-4}$ over 40 epochs per task.
- **Epochs per task:** 40.
- **Batch size:** 32.
- **Seeds:** $\{42, 0, 1, 2, 3\}$ for primary experiments (5 seeds); $\{42, 0, 1\}$ for sweep experiments (3 seeds); $\{42, 0, 1\}$ for TinyImageNet (3 seeds).
- **Data augmentation:** Random horizontal flip and normalisation (CIFAR-10 statistics: mean $[0.491, 0.482, 0.447]$, std $[0.247, 0.243, 0.262]$; TinyImageNet: per-dataset statistics).

B.4 Dataset Details

- **Split-CIFAR-10:** CIFAR-10 [Krizhevsky and Hinton \[2009\]](#) partitioned into 5 sequential two-class tasks: $\{\text{airplane, automobile}\}$, $\{\text{bird, cat}\}$, $\{\text{deer, dog}\}$, $\{\text{frog, horse}\}$, $\{\text{ship, truck}\}$.
- **Split-TinyImageNet:** TinyImageNet [Le and Yang \[2015\]](#) (200 classes, 64×64 images) partitioned into 10 sequential 20-class tasks.

B.5 Hardware and Compute

Experiments were run on a single NVIDIA GeForce GTX 1650 (4 GB VRAM) with CUDA 12.x. Approximate wall-clock training times:

- Split-CIFAR-10, one seed, one method: $\approx 8\text{--}12$ minutes.
- Split-TinyImageNet, one seed, one method: $\approx 35\text{--}50$ minutes.

No multi-GPU or cloud resources were used.

B.6 Code

TCL is implemented in `tar_lab/thermoobserver.py`. The benchmark runner is `tar_lab/continual_benchmark.py`. Experiment configuration uses `tar_lab/schemas.py` (`ContinualLearningBenchmarkConfig`). Result provenance is tracked via an append-only JSONL index and full environment snapshots (Python version, library versions, GPU info, git commit hash).